eCOA Solutions

Database Schema Design

1. Overview

This document defines the database schema for the eCOA Solutions platform. The system uses a hybrid approach with PostgreSQL for relational data (users, studies, submissions) and MongoDB for flexible document storage (form schemas, responses).

1.1 Database Strategy

- **PostgreSQL:** Primary database for structured, relational data
- **MongoDB:** Document store for flexible form schemas and responses
- **Redis:** Session cache and temporary data

1.2 Key Conventions

- All tables use UUID as primary key
- Timestamps use UTC timezone
- Soft deletes via deleted_at column
- Snake_case naming for PostgreSQL tables and columns

2. PostgreSQL Schema

2.1 organizations

Stores sponsor/pharmaceutical company information.

Column	Type	Null	Key	Description
id	UUID	NO	PK	Primary key
name	VARCHAR(255)	NO		Organization name
code	VARCHAR(50)	NO	UQ	Unique org code
type	ENUM	NO		sponsor, cro, site
settings	JSONB	YES		Organization settings
created_at	TIMESTAMP	NO		Creation timestamp
updated_at	TIMESTAMP	NO		Last update
deleted_at	TIMESTAMP	YES		Soft delete

2.2 users

Stores sponsor administrators and other system users.

Column	Type	Null	Key	Description
id	UUID	NO	PK	Primary key
organization_id	UUID	NO	FK	FK to organizations
email	VARCHAR(255)	NO	UQ	Unique email
password_hash	VARCHAR(255)	NO		Bcrypt hash
first_name	VARCHAR(100)	NO		First name
last_name	VARCHAR(100)	NO		Last name
role	ENUM	NO		admin, sponsor, data_mgr
status	ENUM	NO		active, inactive, locked
last_login_at	TIMESTAMP	YES		Last login time
mfa_enabled	BOOLEAN	NO		MFA status
created_at	TIMESTAMP	NO		Creation timestamp
updated_at	TIMESTAMP	NO		Last update

2.3 studies

Stores clinical trial/study metadata.

Column	Type	Null	Key	Description
id	UUID	NO	PK	Primary key
organization_id	UUID	NO	FK	FK to organizations
created_by	UUID	NO	FK	FK to users
name	VARCHAR(255)	NO		Study name
code	VARCHAR(50)	NO	UQ	Unique study code
description	TEXT	YES		Study description
status	ENUM	NO		draft, uat, production
cloned_from	UUID	YES	FK	Source study if cloned
template_id	UUID	YES	FK	FK to templates
settings	JSONB	YES		Study settings
deployed_at	TIMESTAMP	YES		Production deploy time
created_at	TIMESTAMP	NO		Creation timestamp
updated_at	TIMESTAMP	NO		Last update

2.4 forms

Stores form metadata (schema stored in MongoDB).

Column	Type	Null	Key	Description
id	UUID	NO	PK	Primary key
study_id	UUID	NO	FK	FK to studies
name	VARCHAR(255)	NO		Form name
version	INTEGER	NO		Schema version
schema_ref	VARCHAR(50)	NO		MongoDB ObjectId ref
completed_by	ARRAY	NO		patient, caregiver, etc
trigger_type	ENUM	NO		daily, weekly, monthly
trigger_config	JSONB	YES		Trigger settings
notification_config	JSONB	YES		Notification settings
include_in_reports	BOOLEAN	NO		Report inclusion
status	ENUM	NO		draft, active, archived
created_at	TIMESTAMP	NO		Creation timestamp
updated_at	TIMESTAMP	NO		Last update

2.5 templates

Stores reusable form templates.

Column	Type	Null	Key	Description
id	UUID	NO	PK	Primary key
organization_id	UUID	YES	FK	NULL = global template
name	VARCHAR(255)	NO		Template name
type	ENUM	NO		licensed, company, custom
schema_ref	VARCHAR(50)	NO		MongoDB ObjectId ref
license_info	JSONB	YES		License details
is_active	BOOLEAN	NO		Active status
created_at	TIMESTAMP	NO		Creation timestamp

2.6 patients

Stores patient enrollment information.

Column	Type	Null	Key	Description
id	UUID	NO	PK	Primary key
study_id	UUID	NO	FK	FK to studies
patient_code	VARCHAR(50)	NO	UQ	Unique patient code
site_id	VARCHAR(50)	YES		Clinical site ID
enrolled_at	TIMESTAMP	NO		Enrollment date
status	ENUM	NO		active, completed, withdrawn
device_tokens	ARRAY	YES		Push notification tokens
timezone	VARCHAR(50)	YES		Patient timezone
created_at	TIMESTAMP	NO		Creation timestamp

2.7 submissions

Stores form submission metadata (responses in MongoDB).

Column	Type	Null	Key	Description
id	UUID	NO	PK	Primary key
form_id	UUID	NO	FK	FK to forms
patient_id	UUID	NO	FK	FK to patients
response_ref	VARCHAR(50)	NO		MongoDB ObjectId ref
submitted_by	ENUM	NO		patient, caregiver, etc
status	ENUM	NO		pending, completed, partial
started_at	TIMESTAMP	NO		Start time
completed_at	TIMESTAMP	YES		Completion time
device_info	JSONB	YES		Device metadata
created_at	TIMESTAMP	NO		Creation timestamp

2.8 notifications

Stores notification history and scheduling.

Column	Type	Null	Key	Description
id	UUID	NO	PK	Primary key
patient_id	UUID	NO	FK	FK to patients
form_id	UUID	NO	FK	FK to forms
type	ENUM	NO		push, email, sms
status	ENUM	NO		pending, sent, failed
scheduled_at	TIMESTAMP	NO		Scheduled time
sent_at	TIMESTAMP	YES		Actual send time
error_message	TEXT	YES		Error if failed
created_at	TIMESTAMP	NO		Creation timestamp

2.9 audit_logs

Stores audit trail for 21 CFR Part 11 compliance.

Column	Type	Null	Key	Description
id	UUID	NO	PK	Primary key
user_id	UUID	YES	FK	FK to users (NULL if system)
entity_type	VARCHAR(50)	NO		Table/entity name
entity_id	UUID	NO		Entity primary key
action	ENUM	NO		create, update, delete
old_values	JSONB	YES		Previous state
new_values	JSONB	YES		New state
ip_address	INET	YES		Client IP
user_agent	TEXT	YES		Browser/device info
created_at	TIMESTAMP	NO		Event timestamp

3. MongoDB Schema

MongoDB stores flexible document structures for form schemas and responses.

3.1 form_schemas Collection

Stores the complete form definition including pages, elements, and validation rules.

```
{   "_id": ObjectId,   "version": 1,   "pages": [     {       "id": "page-1",
"title": "Page 1",       "elements": [         {           "id": "q1",
"type": "radio",           "label": "How are you feeling today?",
"required": true,           "options": [             { "value": "1", "label": "Very
Poor" },             { "value": "5", "label": "Excellent" }           ],
"properties": {             "font": "Arial",             "size": "medium",
"alignment": "left"           }         }       ]     }   ],   "settings":
{     "showPageNumbers": true,     "allowBackNavigation": true,
"requireReviewPage": true   },   "createdAt": ISODate,   "updatedAt": ISODate }
```

3.2 form_responses Collection

Stores patient form responses with complete answer data.

```
{   "_id": ObjectId,   "submissionId": "uuid-reference",   "schemaVersion": 1,
"answers": {     "q1": {       "value": "4",       "answeredAt": ISODate     },
"q2": {       "value": ["option1", "option3"],       "answeredAt":
ISODate     }   },   "pageProgress": {     "currentPage": 2,     "completedPages":
[1]   },   "metadata": {     "startedAt": ISODate,     "completedAt": ISODate,
"duration": 245   } }
```

4. Entity Relationships

The following describes the key relationships between entities:

4.1 Core Relationships

- **Organization → Users:** One-to-Many (org has many users)
- **Organization → Studies:** One-to-Many (org has many studies)
- **Study → Forms:** One-to-Many (study has many forms)
- **Study → Patients:** One-to-Many (study has many patients)
- **Form → Submissions:** One-to-Many (form has many submissions)
- **Patient → Submissions:** One-to-Many (patient has many submissions)
- **Template → Forms:** One-to-Many (template used by many forms)
- **Study → Study (clone):** Self-referential (study cloned from another)

4.2 Cross-Database References

PostgreSQL tables reference MongoDB documents via string ObjectId references:

- **forms.schema_ref** → MongoDB form_schemas._id
- **templates.schema_ref** → MongoDB form_schemas._id
- **submissions.response_ref** → MongoDB form_responses._id

5. Indexes

5.1 PostgreSQL Indexes

```
-- Users CREATE INDEX idx_users_org ON users(organization_id); CREATE INDEX
idx_users_email ON users(email);  -- Studies   CREATE INDEX idx_studies_org ON
studies(organization_id); CREATE INDEX idx_studies_status ON studies(status);  --
Forms CREATE INDEX idx_forms_study ON forms(study_id); CREATE INDEX
idx_forms_status ON forms(status);  -- Submissions CREATE INDEX
idx_submissions_form ON submissions(form_id); CREATE INDEX idx_submissions_patient
ON submissions(patient_id); CREATE INDEX idx_submissions_status ON
submissions(status);  -- Patients CREATE UNIQUE INDEX idx_patients_code ON
patients(study_id, patient_code);  -- Audit Logs CREATE INDEX idx_audit_entity ON
audit_logs(entity_type, entity_id); CREATE INDEX idx_audit_user ON
audit_logs(user_id); CREATE INDEX idx_audit_created ON audit_logs(created_at);
```

5.2 MongoDB Indexes

```
// form_schemas db.form_schemas.createIndex({ "createdAt": -1 });  //
form_responses db.form_responses.createIndex({ "submissionId": 1 });
db.form_responses.createIndex({ "metadata.completedAt": -1 });
```